Climbing Stairs - Number of Ways Modulo 1e9+7

Problem statement

Given N stairs, count sequences of moves consisting of 1 or 2 stairs that reach exactly N. Output the count modulo 1,000,000,007.

Input format

One integer N on its own line.

Output format

Print the number of ways modulo 1000000007 as a single integer.

Constraints

• 0 <= N <= 1000

Sample input and output

Sample input	Expected output
5	8

Important boundary conditions

There is one empty sequence for N=0. Apply modulo during calculation to avoid overflow.

Implementation expectations

O(N) time and O(1) extra space are possible.

Submission requirements

Submit exactly one .c file. Use stdin and stdout without interactive prompts. Compile with GCC in C11 mode. Avoid undefined behavior, release dynamically allocated memory, and print precisely the required output. Follow the stated input constraints.

Teacher resources (not part of student handout)

A reference oracle, five sample student submissions, assignment-specific rules.json, and 12 verified tests are supplied separately in the instructor test pack. Do not distribute the reference oracle or deliberately faulty variants as student solutions.